ABHIMAN SAKILAM

Email: abhiman.sakilam@gmail.com — **Ph.no**: 8897887693 — **Location**: Hyderabad — **LinkedIn** — **GitHub**

Profile Summary

AI Engineer specializing in LLM and Generative AI systems, with production experience building and deploying RAG pipelines, LLM-powered applications, and fine-tuned VLM/LLM models (QLoRA/PEFT) for real-time structured output.

Experience

- **Detect Technologies** **December 2024 - Present**
  AI Deep Learning Engineer *Chennai, Tamil Nadu*
  - Solely owned and delivered **3 end-to-end industrial AI deployments (Kaneka, UTI Patterson and LMEL)**, developing and deploying **custom safety and security use cases** across challenging environments including **offshore rigs, low-clarity video streams, dynamic FOVs, and varying camera conditions**, while independently handling model integration, optimization, system validation, and production rollout.
  - Engineered and deployed production AI features by **fine-tuning Qwen3-8B VL with QLoRA/PEFT** for pathway, spillage, and trip-hazard detection, leveraging native localization and **XGrammar**-constrained structured outputs to deliver robust **real-time analytics**.
  - Improved PPE compliance analytics by **fine-tuning Vision Transformers (ViT)** with LoRA and deploying **RT-DETR/YOLO26 detectors**, achieving **96% accuracy** and enabling reliable attribute recognition on challenging low-resolution RTSP/RTMP surveillance streams.
  - Architected a scalable **Multi-GPU visual mining platform** integrating zero-shot detection to automate dataset annotation, augmentation, and quality assurance for computer vision training, while reducing raw video storage requirements by **93% (3 TB to 200 GB)** using motion-based filtering and key-frame extraction.
  - Developed a **vehicle–person collision pre-warning system** using **homography-based ground-plane projection** for real-world pedestrian distance estimation, achieving **0.3m error** up to **15m**, reducing false positives by **40%**, and running at above **25 FPS** on live CCTV feeds.
  - Pioneered an **ANPR-based vehicle automation pipeline** with multi-stage **OCR** preprocessing, achieving around **97% plate recognition accuracy at less than 200ms inference latency**, integrated with **PLC-controlled boom barriers** to eliminate manual entry intervention across 2 entry/exit lanes.
  - **Conducted above 50 technical interviews** for AI Software Engineer and AI/ML Intern roles, evaluating candidates across Machine Learning, Deep Learning, Computer Vision, and NLP.

Education

- **Amrita Vishwa Vidyapeetam** **August 2020 - April 2024**
  B.Tech in Artificial Intelligence *Chennai, Tamil Nadu*

Projects

- **MeetMind : AI Assisted Meeting Platform** **Sep 2025 – Oct 2025**
  Personal Project
  - **Multi-Agent Orchestration**: LangChain-powered intent router dispatches live meeting audio to **7 specialized agents** (summarize, action items, decisions, screen analysis) in real-time, **Redis** write-through persistence per session
  - GPU-accelerated local STT pipeline : **Distil-Whisper** on **CUDA** via **faster-Whisper**, streamed over **WebSockets** into FastAPI backend, achieving low-latency transcription without cloud STT costs.
  - Built a **full-stack meeting intelligence platform** with **WebRTC (PeerJS) P2P video**, **JWT auth**, and automated AI summary generation, PDF export, and SMTP delivery in **<10s post-meeting**; **Dockerized** with GPU support enabling **single-command deployment** and reducing setup time from hours to **<2 minutes**.

- **RAG-Based Biology and Medical Q/A System** **Feb 2025 – Mar 2025**
  Personal Project
  - Built a production RAG system for biomedical Q&A over a 40K-passage corpus, implementing a multi-stage retrieval pipeline - **BM25 sparse retrieval, FAISS dense vector search, Reciprocal Rank Fusion, and neural cross-encoder reranking** - behind independently toggleable config flags for per-component ablation.
  - Designed an evaluation harness measuring Recall/MRR/nDCG@k against 707 gold-labeled QA pairs **(BM25 baseline: MRR@5 0.83, nDCG@5 0.74)**, sharing one retrieval code path between production and eval to eliminate train/serve skew; added H**yDE query rewriting, citation validation, and an LLM-judge groundedness guardrail.**

- Shipped the full stack **FastAPI backend, React/Vite UI, multi-stage Docker build** with 147 tests and layered graceful degradation, so reranker, rewrite, or guardrail failures never drop a user request.

- **Automated Prompt Optimizer for VLMs** **Feb 2025 – Mar 2025**
  Detect Technologies
  - Developed a **production-grade iterative prompt tuning pipeline** for VLMs (**Qwen3-VL, InternVL3.5, Molmo**) with ground-truth validation and FP/FN-based evaluation to autonomously identify classification failures and refine prompts.
  - Reduced **prompt engineering effort by 50%** and accelerated **global use-case deployment timelines by 80%** through automated, self-corrective optimization.

Skills

- **AI & Machine Learning**: Generative AI, Agentic AI, MCP, LLMs, Prompt Engineering, LoRA/QLoRA, PEFT, Quantization, Inference Optimization, Transfer Learning, Model Evaluation, Pinecone, FAISS, VectorDBs, Prompt engineering, RAG, Context Engineering/Context Management, OpenAI
- **Programming**: Python, Java, MySQL, DSA, OOPs
- **Frameworks & Libraries**: PyTorch, TensorFlow, Hugging Face, OpenCV, LangChain, LangGraph, Keras, FastAPI
- **Systems & Deployment**: System Design, Docker, Linux, Redis, AWS (Bedrock, EC2, S3), GitLab CI/CD, NVIDIA DeepStream, RESTful APIs, PostgreSQL, JIRA
- **Coding Assistants**: Cursor, Claude Code

Publications

- **Towards Autonomous Driving System Using Behavioral Cloning Approach** 🔗: **IEEE Conference – Aug 2023**
- **A Comparative Analysis of Satellite–Map Image Translation Using GANs** 🔗: **IEEE Conference – May 2023**